MOTOR VIBRATION SIGNAL ANALYSIS

ADXL345 + ESP32 Experimental Measurement

Item	Value
Dataset	datasuddenloas2\processed\adxl345_5sec_calibrated.csv
Date	2026-08-20 22:14
Sampling Frequency	1559.000 Hz
Samples	4.994e+04
Duration	32.0327 s

1. Objective

The objective of this script is to analyze experimentally measured motor vibration data, extract time-domain and frequency-domain features, visualize the measured acceleration signals, and prepare a professional engineering report. The analysis is descriptive and comparative; it does not classify the motor condition or automatically diagnose faults.

2. Experimental Data

Parameter	Value
Input file	datasuddenloas2\processed\adxl345_5sec_calibrated.csv
Detected X column	X_g
Detected Y column	Y_g
Detected Z column	Z_g
Acceleration units	g
Sampling frequency source	Configured FS parameter in analysis.py
Sampling frequency	1559.000 Hz
Number of samples	4.994e+04
Duration	32.032713 s
Nyquist frequency	779.500 Hz
Frequency resolution	0.031218 Hz/bin

3. Signal Processing

The input data are already calibrated acceleration values in g, so this script does not perform sensor calibration. The calibrated signals are preserved. For vibration feature extraction and FFT, the remaining DC component is removed using x_ac = x - mean(x), y_ac = y - mean(y), and z_ac = z - mean(z). The resultant signal is R = sqrt(X^2 + Y^2 + Z^2), representing the magnitude of the three-axis acceleration vector; its AC vibration component is obtained by removing its mean. FFT spectra are calculated from DC-removed signals using a Hann window to reduce spectral leakage. Frequency resolution is FS/N.

4. Time-Domain Analysis

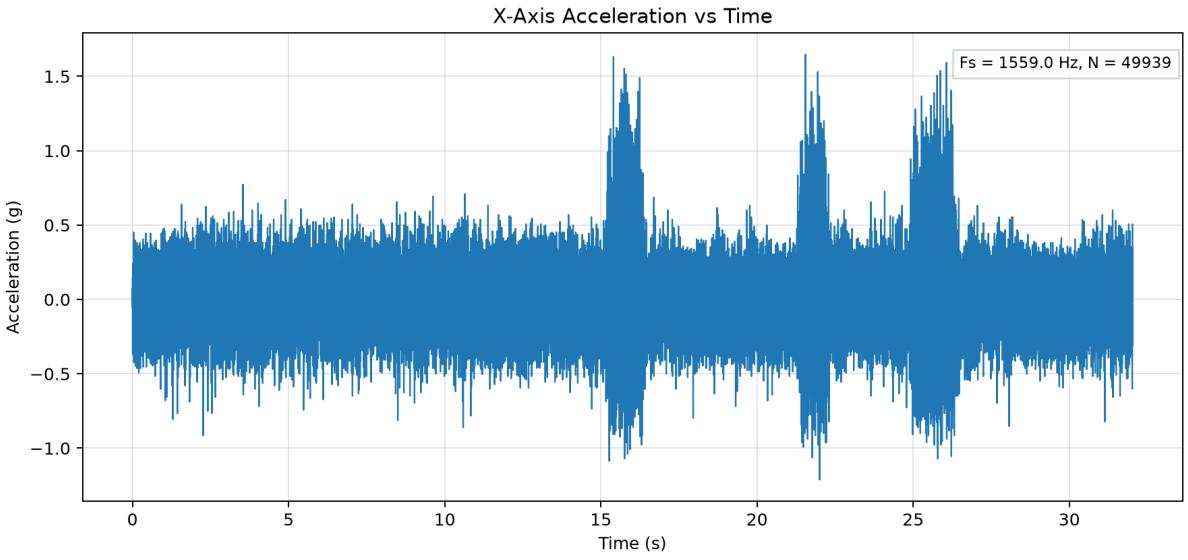


Figure 1. X-axis calibrated acceleration versus time.

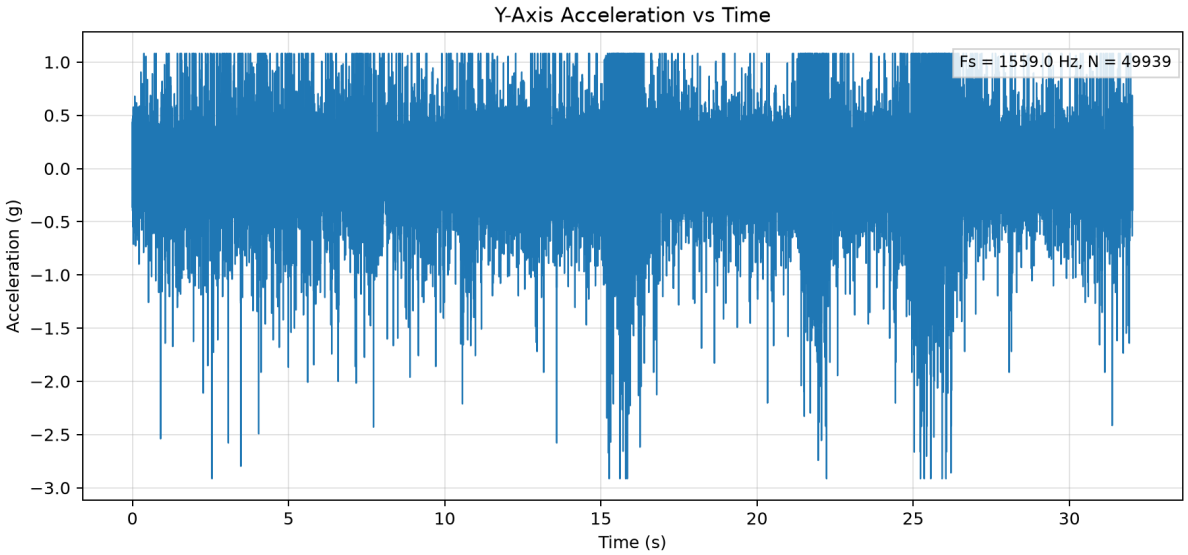


Figure 2. Y-axis calibrated acceleration versus time.

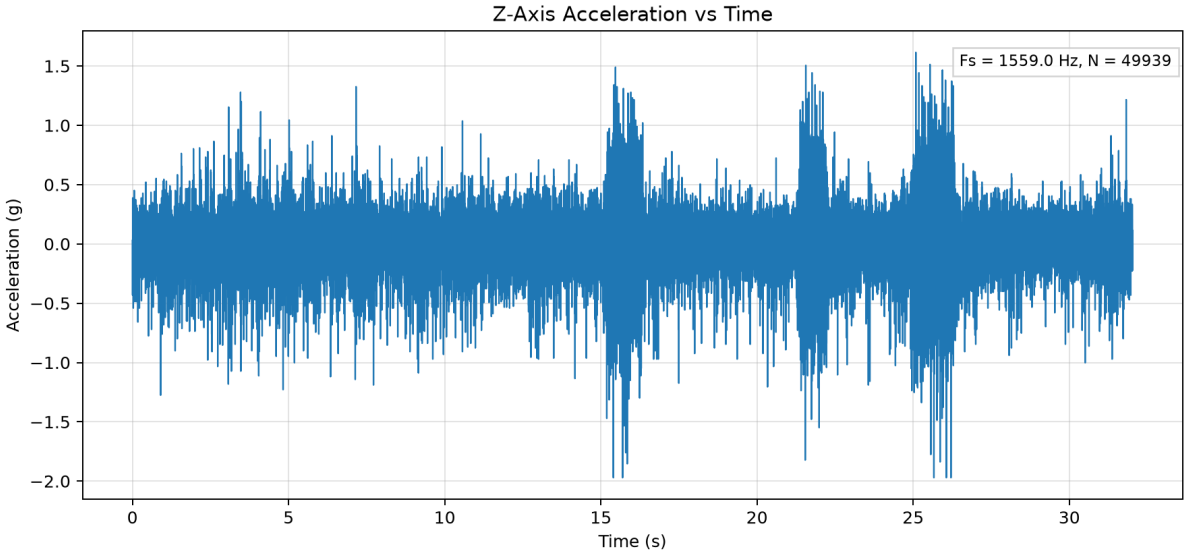


Figure 3. Z-axis calibrated acceleration versus time.

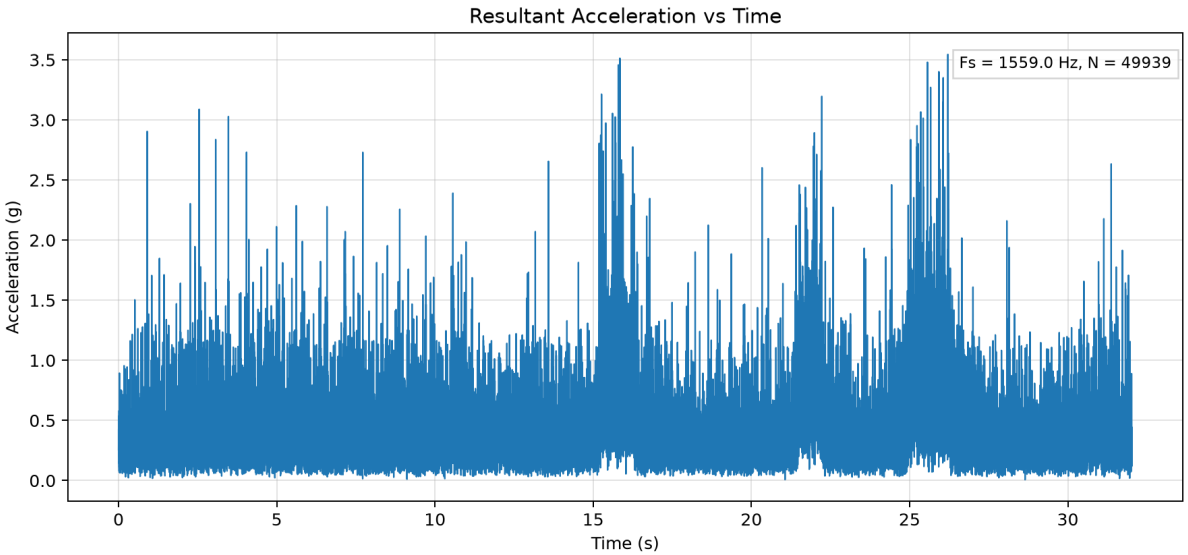


Figure 4. Resultant calibrated acceleration magnitude versus time.

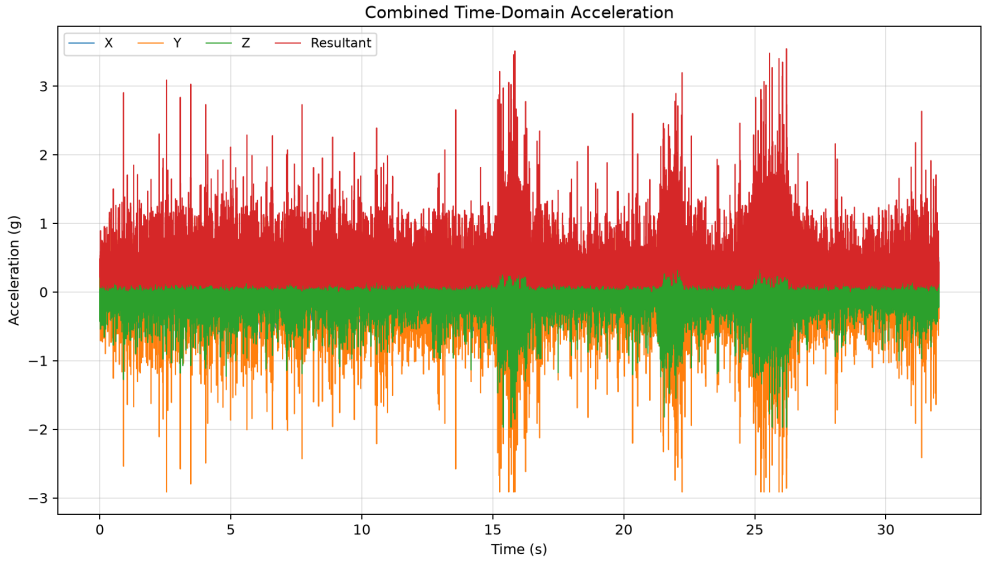


Figure 5. Combined time-domain acceleration comparison.

Table 1. Time-domain vibration features calculated from DC-removed AC signals.

Feature	X	Y	Z	Resultant
Mean	6.011e-18	-1.209e-18	6.972e-18	-1.138e-17
Median	0.00635	0.00826	0.03306	-0.08072
Standard Deviation	0.23134	0.40203	0.23845	0.30467
Variance	0.05352	0.16163	0.05686	0.09282
RMS	0.23134	0.40203	0.23844	0.30466
Maximum	1.64697	1.09030	1.61900	3.12202
Minimum	-1.21240	-2.90580	-1.96694	-0.41609
Peak	1.64697	2.90580	1.96694	3.12202
Absolute Peak	1.64697	2.90580	1.96694	3.12202
Peak-to-Peak	2.85938	3.99609	3.58594	3.53811
Range	2.85938	3.99609	3.58594	3.53811
Crest Factor	7.11929	7.22786	8.24905	10.24743
Shape Factor	1.32963	1.40910	1.43876	1.43907
Impulse Factor	9.46602	10.18475	11.86838	14.74676
Clearance Factor	11.42114	12.69200	14.84156	18.02997
Skewness	0.18401	-0.62647	-0.36539	2.37821
Kurtosis	2.37872	4.22287	4.83173	9.48302
Mean Absolute Value	0.17399	0.28531	0.16573	0.21171
Energy	2672.62985	8071.43545	2839.30331	4635.33361
Signal Power	0.05352	0.16163	0.05686	0.09282
Median Absolute Deviation	0.14062	0.20312	0.11719	0.13106
Q1	-0.14209	-0.19486	-0.10756	-0.19390
Q2 / Median	0.00635	0.00826	0.03306	-0.08072
Q3	0.13916	0.21139	0.11900	0.08808
Interquartile Range	0.28125	0.40625	0.22656	0.28198
Zero-Crossing Rate	0.32947	0.34102	0.32993	0.29428

5. Time-Domain Feature Definitions

Feature	Definition	Engineering Meaning
Mean	mean(x)	Average acceleration level. For AC vibration it should be close to zero after DC removal.
Median	middle value of x	Robust central value that is less sensitive to isolated peaks than the mean.
Standard Deviation	sqrt(sample variance)	Spread of vibration amplitude around the mean; higher values indicate greater variation.
Variance	sum((x-mean)^2)/(N-1)	Mean-square spread of the signal; useful for comparing vibration intensity.
RMS	sqrt(mean(x^2))	Overall vibration intensity or energy level of the acceleration signal.
Peak	max(abs(x))	Largest absolute acceleration excursion recorded in the signal.
Peak-to-Peak	max(x)-min(x)	Total excursion range between positive and negative extremes.
Crest Factor	peak/RMS	Peak severity relative to average vibration level; higher values may indicate impulsive behavior.
Shape Factor	RMS/mean(abs(x))	Describes waveform shape independent of absolute amplitude.
Impulse Factor	peak/mean(abs(x))	Highlights isolated high-amplitude events relative to average absolute motion.
Clearance Factor	peak/(mean(sqrt(abs(x)))^2)	Sensitive to sharp peaks and often used when inspecting impulsive vibration characteristics.
Skewness	third standardized moment	Measures asymmetry of amplitude distribution.
Kurtosis	excess fourth standardized moment	Measures heaviness of tails and peakedness; larger values can be associated with impulsive events.
Energy	sum(x^2)	Total discrete signal energy over the acquisition duration.
Signal Power	mean(x^2)	Average energy per sample; equal to RMS squared.
Zero-Crossing Rate	fraction of sign changes	Indicates how often the AC signal crosses zero and can reflect frequency content.

6. Frequency-Domain Analysis

Single-sided amplitude spectra are computed from DC-removed signals. The frequency associated with an FFT bin is a spectral component frequency, not automatically the motor shaft speed.

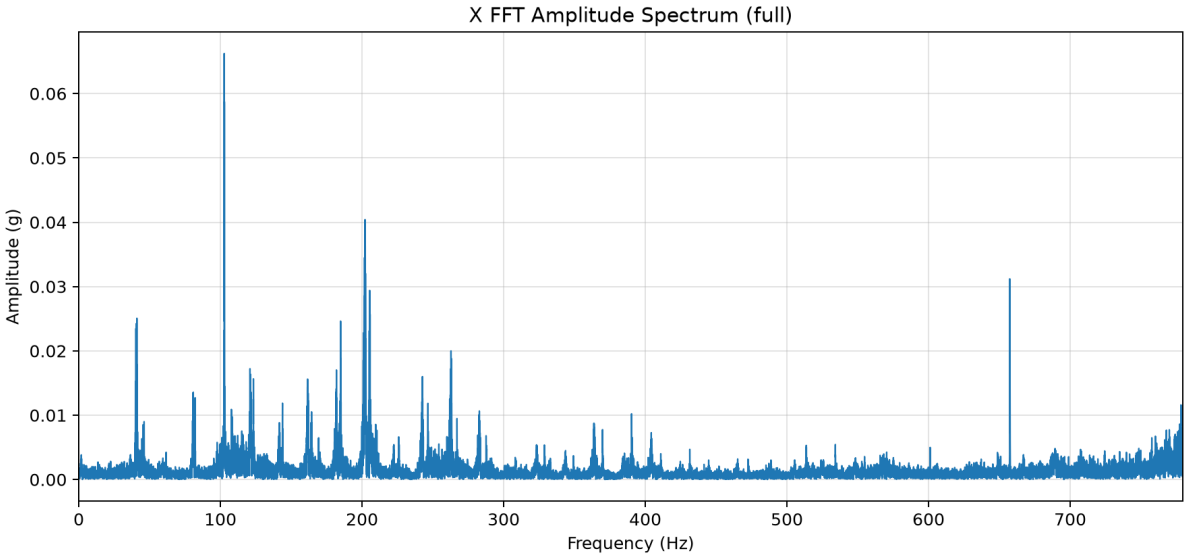


Figure 6. X-axis full FFT spectrum.

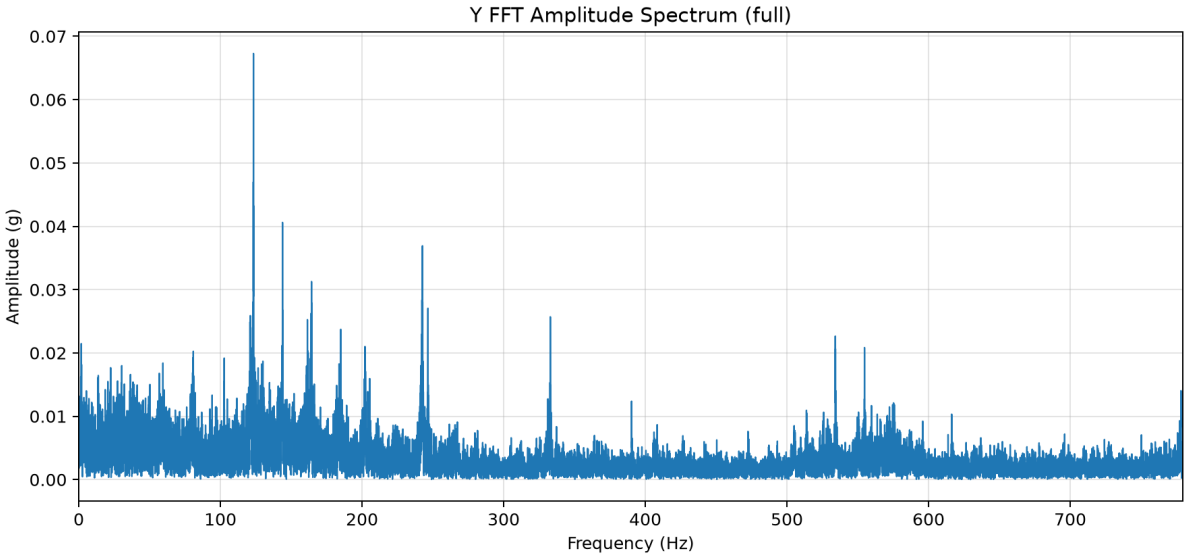


Figure 7. Y-axis full FFT spectrum.

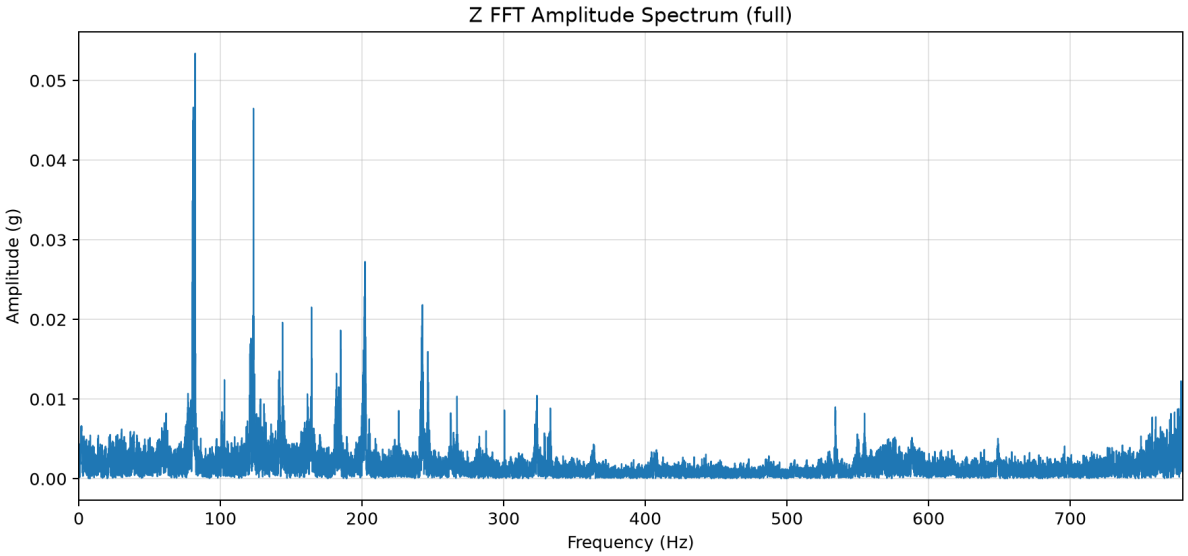


Figure 8. Z-axis full FFT spectrum.

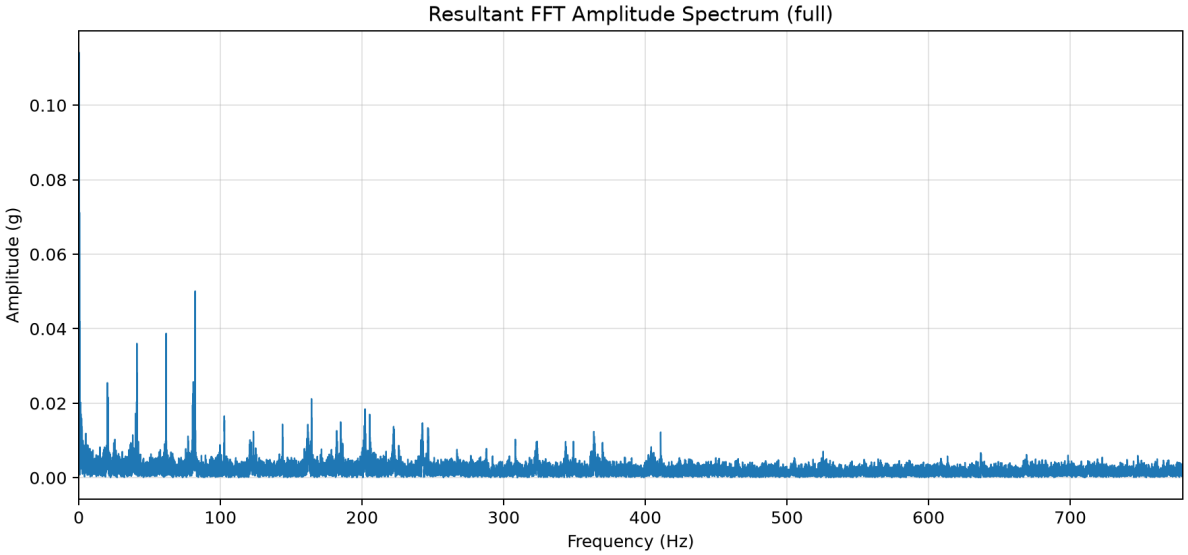


Figure 9. Resultant full FFT spectrum.

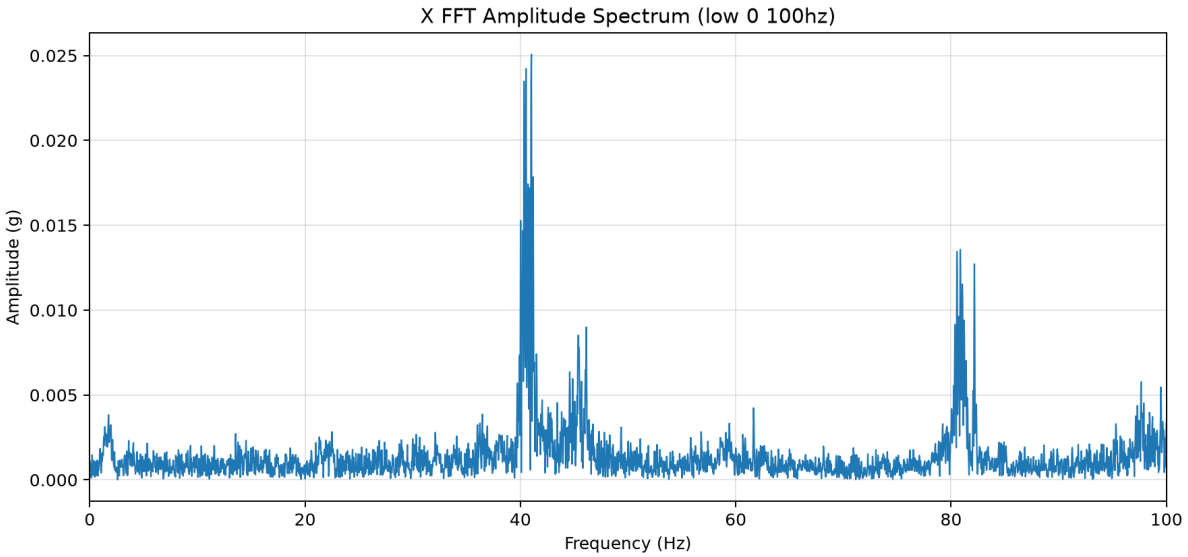


Figure 10. X-axis low-frequency FFT spectrum, 0-100 Hz.

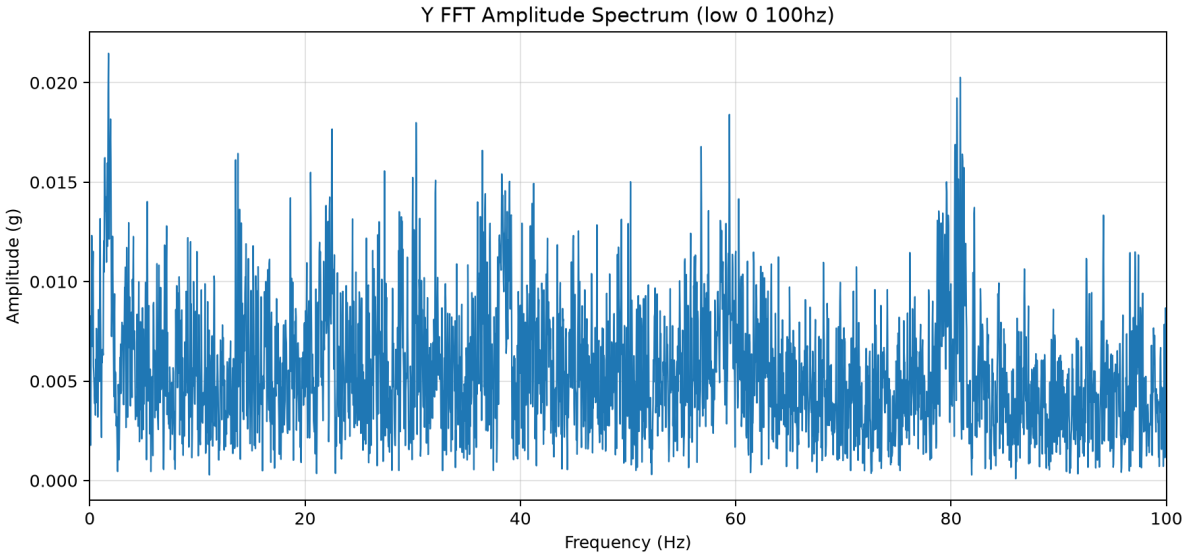


Figure 11. Y-axis low-frequency FFT spectrum, 0-100 Hz.

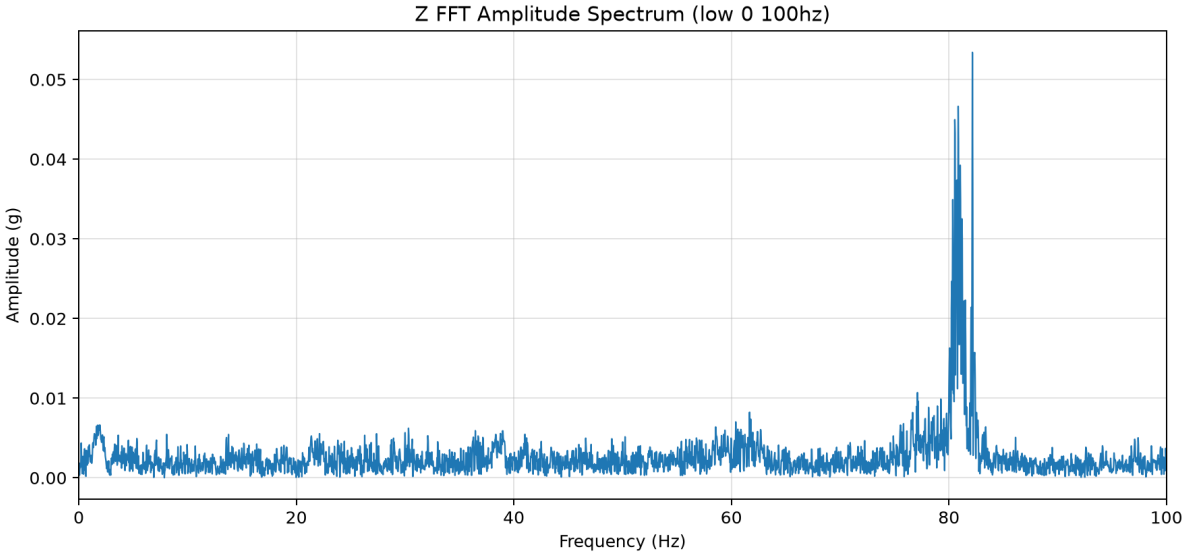


Figure 12. Z-axis low-frequency FFT spectrum, 0-100 Hz.

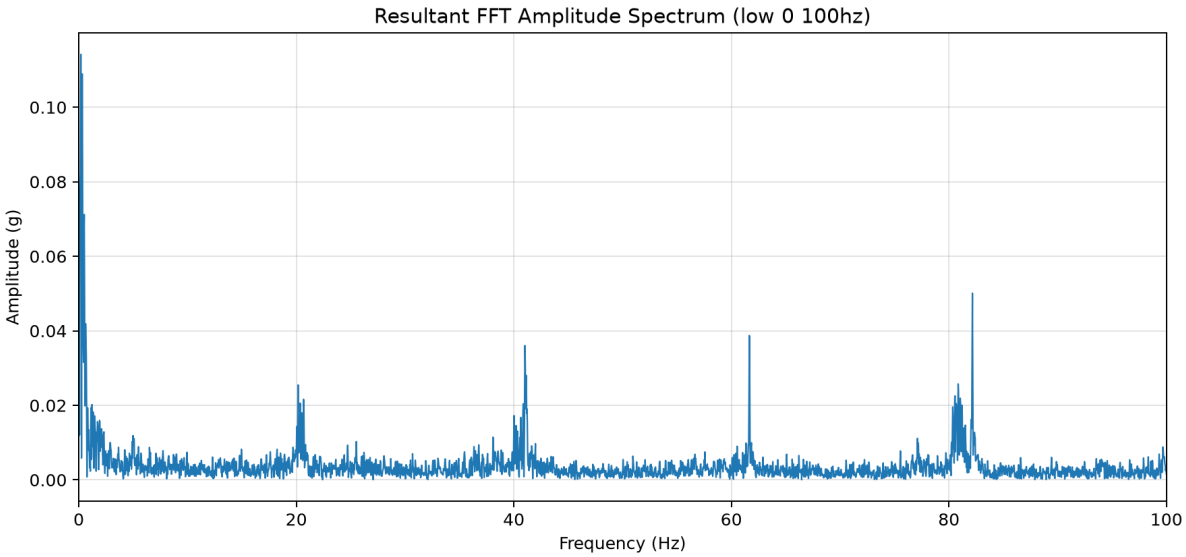


Figure 13. Resultant low-frequency FFT spectrum, 0-100 Hz.

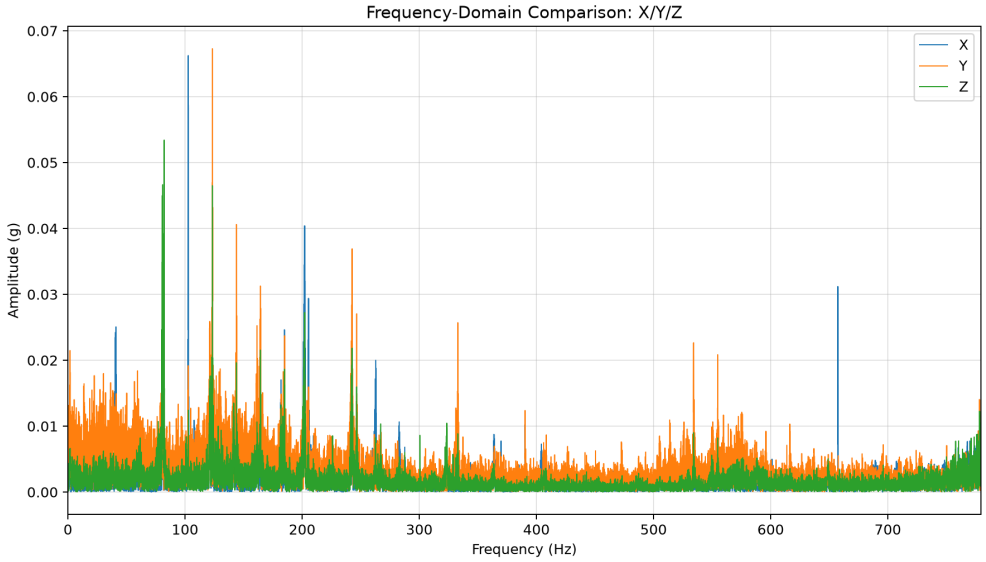


Figure 14. Full-spectrum comparison of X/Y/Z axes.

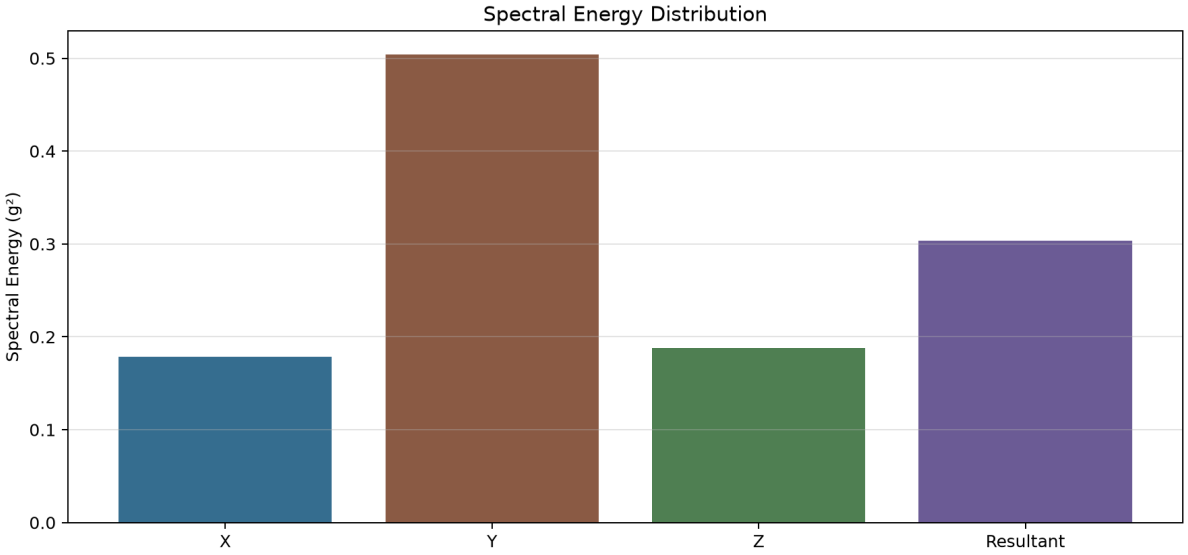


Figure 15. Spectral energy distribution comparison.

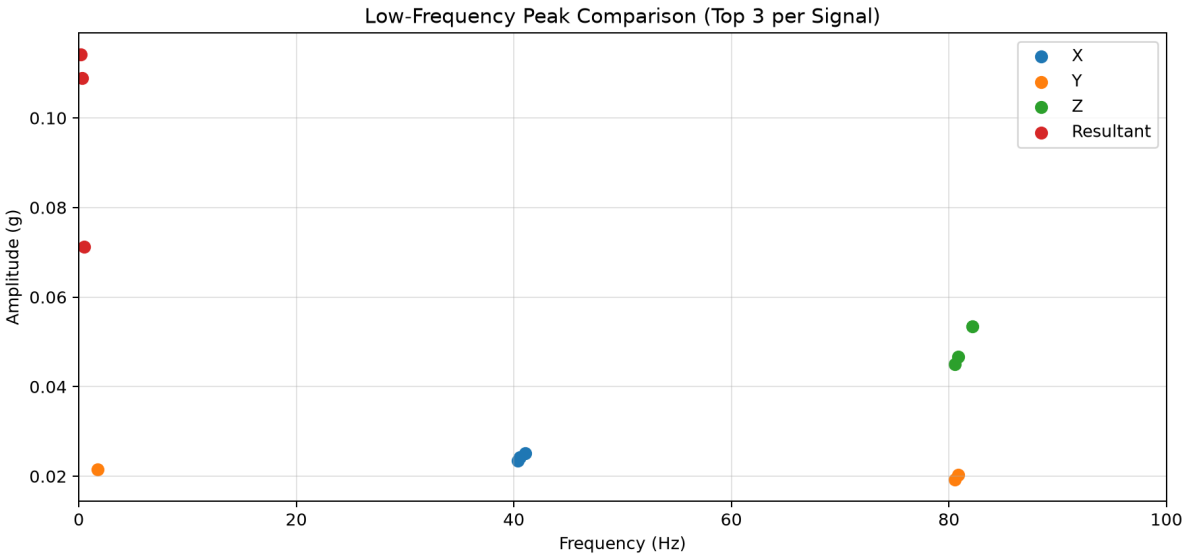


Figure 16. Low-frequency peak comparison.

7. Frequency-Domain Features

Table 2. Frequency-domain features calculated from Hann-windowed AC signals.

Feature	X	Y	Z	Resultant
Dominant Frequency	102.67629	123.34266	82.16600	0.18731
Dominant Amplitude	0.06623	0.06728	0.05341	0.11418
Spectral Centroid	352.40556	305.43127	327.16403	324.58826
Spectral RMS Frequency	325.65049	299.04426	310.52699	271.90328
Spectral Bandwidth	224.45773	212.44978	230.37498	258.82533
Spectral Energy	0.17884	0.50402	0.18770	0.30384
Spectral Variance	5.038e+04	4.513e+04	5.307e+04	6.699e+04
Spectral Skewness	-0.07837	-0.08676	-0.00577	-0.68649
Spectral Kurtosis	1.72310	1.79249	1.74850	1.44317
Spectral Flatness	0.67062	0.73375	0.70667	0.76816
Spectral Entropy	0.75468	0.88338	0.80276	0.76333
Spectral Roll-off	514.47406	520.87377	531.01964	406.86532
Spectral Peak	0.06623	0.06728	0.05341	0.11418
Number of Significant Peaks	993.00000	3148.00000	1930.00000	855.00000

Feature	Definition	Engineering Meaning
Dominant Frequency	frequency of largest non-DC FFT amplitude	Strongest periodic component in the measured vibration spectrum.
Dominant Amplitude	largest non-DC FFT amplitude	Amplitude of the strongest spectral component in g.
Spectral Centroid	sum(f*A)/sum(A)	Amplitude-weighted center frequency; higher values indicate energy shifted toward higher frequencies.
Spectral RMS Frequency	sqrt(sum(f^2*P)/sum(P))	Power-weighted frequency magnitude of the spectrum.
Spectral Bandwidth	sqrt(sum((f-centroid)^2*P)/sum(P))	Width of spectral energy around the centroid.
Spectral Energy	sum(A^2)	Total spectrum energy measure from FFT amplitudes.
Spectral Flatness	geometric mean(A)/arithmetic mean(A)	Indicates whether the spectrum is tone-like or broadly distributed.
Spectral Entropy	normalized entropy of spectral power	Measures how concentrated or spread out the spectral power is.
Spectral Roll-off	frequency below which 85% of power lies	Shows the frequency range containing most spectral energy.

8. Spectral Peak Analysis

Equivalent RPM is calculated as frequency x 60 and is labeled only as the equivalent rotational speed if that frequency corresponds to 1x shaft rotational frequency. The script does not assert that any component is definitely shaft speed.

Table 3. Most significant full-spectrum peaks.

Axis	Rank	Frequency (Hz)	Amplitude (g)	Equivalent RPM
X	1.0000	102.6763	0.0662	6160.5771
X	2.0000	102.8324	0.0587	6169.9425
X	3.0000	202.1059	0.0404	1.213e+04
X	4.0000	201.9186	0.0381	1.212e+04
X	5.0000	202.2620	0.0367	1.214e+04
X	6.0000	201.7625	0.0345	1.211e+04
X	7.0000	202.4181	0.0320	1.215e+04
X	8.0000	657.3280	0.0312	3.944e+04
X	9.0000	205.4150	0.0294	1.232e+04
X	10.0000	201.6376	0.0285	1.210e+04
Y	1.0000	123.3427	0.0673	7400.5595
Y	2.0000	123.2178	0.0470	7393.0671
Y	3.0000	123.4675	0.0432	7408.0518
Y	4.0000	143.8529	0.0406	8631.1764
Y	5.0000	242.6270	0.0369	1.456e+04
Y	6.0000	164.3320	0.0313	9859.9203
Y	7.0000	242.2836	0.0284	1.454e+04
Y	8.0000	242.7831	0.0280	1.457e+04
Y	9.0000	122.9993	0.0280	7379.9555
Y	10.0000	164.5193	0.0279	9871.1588
Z	1.0000	82.1660	0.0534	4929.9602
Z	2.0000	80.8548	0.0466	4851.2906
Z	3.0000	123.3427	0.0465	7400.5595
Z	4.0000	80.5427	0.0450	4832.5597
Z	5.0000	81.0422	0.0392	4862.5291
Z	6.0000	80.7300	0.0374	4843.7982
Z	7.0000	80.6675	0.0354	4840.0521
Z	8.0000	80.3554	0.0349	4821.3212
Z	9.0000	81.1982	0.0325	4871.8945
Z	10.0000	202.1059	0.0273	1.213e+04
Resultant	1.0000	0.1873	0.1142	11.2385
Resultant	2.0000	0.3122	0.1090	18.7309
Resultant	3.0000	0.4995	0.0712	29.9694
Resultant	4.0000	82.1660	0.0501	4929.9602
Resultant	5.0000	0.6244	0.0419	37.4617
Resultant	6.0000	61.6557	0.0387	3699.3432

Axis	Rank	Frequency (Hz)	Amplitude (g)	Equivalent RPM
Resultant	7.0000	41.0206	0.0360	2461.2339
Resultant	8.0000	41.1142	0.0281	2466.8532
Resultant	9.0000	80.8548	0.0257	4851.2906
Resultant	10.0000	20.1669	0.0255	1210.0130

Table 4. Most significant low-frequency peaks from 0-100 Hz.

Axis	Rank	Frequency (Hz)	Amplitude (g)	Equivalent RPM
X	1.0000	41.0206	0.0251	2461.2339
X	2.0000	40.5211	0.0242	2431.2645
X	3.0000	40.3338	0.0235	2420.0260
X	4.0000	41.1767	0.0178	2470.5993
X	5.0000	40.7084	0.0174	2442.5031
X	6.0000	40.8333	0.0172	2449.9954
X	7.0000	40.0216	0.0153	2401.2952
X	8.0000	40.2401	0.0147	2414.4068
X	9.0000	80.8548	0.0136	4851.2906
X	10.0000	80.5427	0.0135	4832.5597
Y	1.0000	1.7482	0.0215	104.8928
Y	2.0000	80.8548	0.0203	4851.2906
Y	3.0000	80.5427	0.0192	4832.5597
Y	4.0000	59.4080	0.0184	3564.4811
Y	5.0000	1.9355	0.0182	116.1313
Y	6.0000	30.3128	0.0180	1818.7657
Y	7.0000	22.5082	0.0177	1350.4944
Y	8.0000	80.3866	0.0169	4823.1943
Y	9.0000	56.7857	0.0168	3407.1419
Y	10.0000	36.4627	0.0166	2187.7635
Z	1.0000	82.1660	0.0534	4929.9602
Z	2.0000	80.8548	0.0466	4851.2906
Z	3.0000	80.5427	0.0450	4832.5597
Z	4.0000	81.0422	0.0392	4862.5291
Z	5.0000	80.7300	0.0374	4843.7982
Z	6.0000	80.6675	0.0354	4840.0521
Z	7.0000	80.3554	0.0349	4821.3212
Z	8.0000	81.1982	0.0325	4871.8945
Z	9.0000	80.2617	0.0247	4815.7020
Z	10.0000	81.5104	0.0223	4890.6254
Resultant	1.0000	0.1873	0.1142	11.2385
Resultant	2.0000	0.3122	0.1090	18.7309
Resultant	3.0000	0.4995	0.0712	29.9694
Resultant	4.0000	82.1660	0.0501	4929.9602

Axis	Rank	Frequency (Hz)	Amplitude (g)	Equivalent RPM
Resultant	5.0000	0.6244	0.0419	37.4617
Resultant	6.0000	61.6557	0.0387	3699.3432
Resultant	7.0000	41.0206	0.0360	2461.2339
Resultant	8.0000	41.1142	0.0281	2466.8532
Resultant	9.0000	80.8548	0.0257	4851.2906
Resultant	10.0000	20.1669	0.0255	1210.0130

9. Feature Interpretation

RMS represents the overall vibration energy/intensity of the measured acceleration signal. An increase generally indicates stronger vibration under the tested condition.

Peak and peak-to-peak values describe the largest acceleration excursions. Larger values may indicate stronger transient or periodic vibration, but they should be compared across controlled experiments.

Crest factor, impulse factor, clearance factor, and kurtosis emphasize impulsive behavior. Higher values can be associated with sharper events, but none of these features alone proves a fault.

Dominant frequency is the largest non-DC spectral component. It identifies the strongest periodic content in the measured signal.

Spectral centroid and RMS frequency indicate where spectral content is concentrated. Higher values generally mean energy has shifted toward higher frequencies.

Spectral bandwidth, entropy, and flatness describe how concentrated or spread out the spectrum is. These features are useful for comparing operating or physical conditions.

Low-frequency peaks in the 0-100 Hz range are separated because motor rotational and harmonic components often appear in this region, depending on the experimental setup.

10. Observed Vibration Characteristics

The highest RMS vibration level is observed in Y with RMS = 0.4020 g.

The largest absolute peak is observed in Resultant with peak = 3.1220 g.

The strongest dominant spectral amplitude is observed in Resultant: 0.1873 Hz at 0.1142 g.

The strongest component in the 0-100 Hz band is on Resultant at 0.1873 Hz with amplitude 0.1142 g.

The highest crest factor is on Resultant (10.2474), which may indicate stronger transient content and should be compared with other controlled conditions.

The highest kurtosis is on Resultant (9.4830), which provides an indication of distribution peakedness but is not a fault diagnosis by itself.

11. Conclusion

This report summarizes measured vibration characteristics from the provided calibrated ADXL345 dataset. The extracted features and plots provide a basis for comparing this condition against other experimentally controlled motor conditions. No automatic health/fault label is assigned.

12. Appendix

Processing Parameters

Parameter	Value
FFT window	Hann
Low-frequency band	0-100.0 Hz
Top peaks per signal	10.0000
Peak prominence ratio	0.0300
Spectral roll-off percentage	0.8500
Input file	datasuddenloas2\processed\adxl345_5sec_calibrated.csv
Output results folder	datasuddenloas2\results

Complete feature tables and peak tables are saved as CSV files in data/results.